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Design and Reproducible Software-in-the-Loop Validation of HXNS

A Hybrid Optical-Pulsar-Inertial Navigation Architecture for Deep-Space Spacecraft

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Abstract. HXNS is a Rust no_std deep-space navigation prototype that fuses calibrated optical range and bearing, inertial propagation, processed pulsar time-of-arrival products, atomic-clock state, and packed planetary ephemerides. The flight-oriented crates are separated from a host simulator but are exercised by the same software-in-the-loop path. Version 0.2 documents a deterministic 640-run paired-seed campaign spanning two sensor profiles, four mission windows, and four sensor suites. All 640 runs were finite and valid. The results show scenario-dependent behaviour: pulsar fusion lowered run-mean error in 155 of 160 paired comparisons with IMU-plus-optical, while worst-error values were unchanged in 119 pairs because shared fault peaks often dominated. A counterintuitive cruise result exposes an optical-model/filter mismatch that remains unresolved. HXNS is therefore presented as a self-assessed TRL 4 research prototype and evaluation environment, not a flight-qualified navigation computer or validated Neptune mission design.

Keywords: deep-space navigation; XNAV; optical navigation; inertial navigation; extended Kalman filter; Rust; no_std; software-in-the-loop; paired-seed validation

Scope and disclosure. All quantitative results are conditional on synthetic sensor assumptions, a shared project truth model, bounded 120-second windows, and the documented fault schedule. They are not camera or pulsar hardware calibration, independent orbit-determination validation, LEON5 timing evidence, radiation qualification, launch approval, or operational accuracy for a real spacecraft.

1. Motivation

Deep-space spacecraft benefit from highly capable Earth-based tracking, but communication delay, limited tracking time, and interrupted contact motivate useful onboard autonomy. Optical navigation provides geometric information from solar-system bodies, inertial sensors bridge intervals between external observations, and pulsar timing provides a directionally distinct timing observable. These products have different cadences and failure modes; integrating them is therefore a problem in coordinate frames, time correlation, calibration, uncertainty, and fault handling rather than simply intersecting idealized spheres.

HXNS was built to make that integration inspectable at an embedded-oriented software boundary. A host simulator supplies truth, synthetic noise, and faults, while flight-oriented Rust crates consume typed products and produce a navigation solution. This separation makes it possible to test the intended estimator rather than a substitute hidden inside the simulator, and provides a defined path toward later hardware-in-the-loop work.

2. System architecture and interfaces

The flight core consumes processed sensor products rather than camera pixels or raw X-ray photon events. A camera front end supplies body identity, centroid, apparent diameter, quality, and calibration metadata. The shared flight path converts angular diameter and a body-diameter model into range, and rotates a calibrated body-frame line of sight into ICRF. A pulsar receiver supplies one processed TOA residual and uncertainty after an integration window; raw pulse count remains diagnostic and is never treated as multiple independent EKF measurements.

Navigation state is expressed in the Solar-System Barycentric frame using ICRF/J2000 axes. The atomic clock is correlated to TDB through a launch reference, tick rate, bias, and drift; a CPU wall clock is not used as navigation time. Planetary states are evaluated from a compact DE442-derived Chebyshev pack held in read-only memory, keeping raw JPL ASCII parsing outside the flight image.

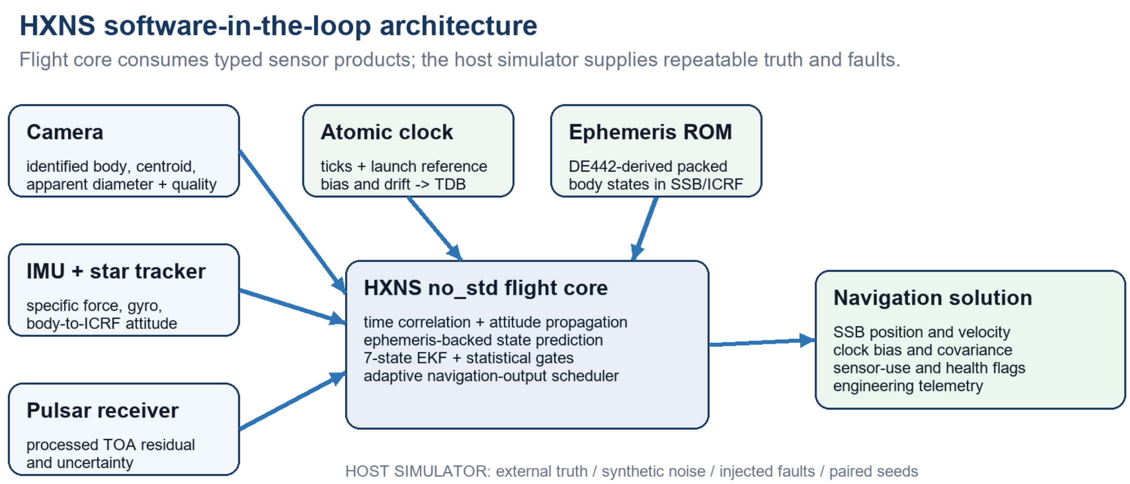

Figure 1. HXNS information flow. The host supplies truth, noise, and faults; the no_std flight boundary consumes typed products and emits navigation state and telemetry.

3. Estimation and asynchronous scheduling

The current seven-state extended Kalman filter estimates three-dimensional position, velocity, and clock bias. Prediction integrates inertial specific force after attitude rotation and adds ephemeris-backed point-mass gravity. Sequential measurement updates accept optical range, two tangent-plane bearing components, and pulsar timing residuals when normalized innovations pass a six-sigma gate. Joseph-form covariance updates preserve numerical symmetry and robustness. Attitude estimation remains separate: gyro propagation is corrected by star-tracker observations before camera geometry is fused.

Navigation-output cadence is selected from the nearest configured reference-body distance: 10 Hz below 50,000 km, 5 Hz from 50,000 to 75,000 km, and 1 Hz beyond 75,000 km. This is not pulsar cadence. On navigation ticks without a fresh camera or TOA product, the EKF predicts but does not reuse stale measurements.

Table 1. Default independent product schedules and flight-core treatment.

Product	Default cadence	Current prototype treatment
IMU	100 Hz	Integrated between navigation outputs.
Camera set	1 Hz	Targeted Earth/Jupiter/Neptune attempts; visibility and uncertainty propagated.
Processed pulsar TOA	Every 10 s	One residual after a 10 s integration window; raw pulses diagnostic.
Navigation output	1/5/10 Hz	Truth-distance-selected in the host campaign; independent of sensor cadence.
Ephemeris	On demand	DE442-derived compact pack in SSB ICRF/J2000 context.

4. Reproducible validation campaign

The validation runner loads the DE442 pack and screened Earth-Jupiter-Neptune trajectory once, then caches each exact adaptive-rate truth window. Every sensor suite within a scenario, profile, and seed reuses mission truth, initialization, covariance, cadence, fault windows, and sensor-generation order. Products are generated even when a suite disables their fusion, preserving common random numbers and allowing paired comparisons.

The campaign combines 20 deterministic seeds, two engineering sensor profiles, four mission scenarios, and four sensor suites (IMU-only, IMU-plus-optical, IMU-plus-pulsar, and full hybrid), producing 640 runs. Each run covers 120 seconds. The report records mean, P50, P95, worst, and final position error; measurement outcomes; adaptive-rate counts; non-finite samples; and fault containment or recovery. All 640 runs were valid with zero non-finite estimator samples. The Rust workspace also passed 74 automated tests on 8 August 2026.

Table 2. Campaign matrix reproduced from the generated validation report.

Dimension	Values	Count
Seeds	20 deterministic paired values starting at 0x48584e5350484153	20
Profiles	Nominal; degraded	2
Scenarios	Near Earth; Earth transition; deep-space cruise; Neptune capture	4
Sensor suites	IMU-only; IMU+optical; IMU+pulsar; full hybrid	4
Total runs	20 x 2 x 4 x 4	640
Valid / invalid	Finite and structurally valid / invalid	640 / 0

Provenance. Source report: assets/generated/hxns_validation_report.json; HXNS version 0.1.0; generated 1 August 2026; SHA-256 87CF460EC10A137A728840F61DE8D00E4A2F659180E2347C8E178825D084D1A1.

5. Quantitative results

Table 3 reports full-hybrid aggregates over 20 seeds per scenario and profile. 'Mean' is the mean of each run's mean position error; 'P95' is the 95th percentile across run-level P95 values; 'worst' is the maximum observed sample; and 'final' is mean final position error. Units are kilometres. These numbers characterize the bounded synthetic campaign only.

Table 3. Full-hybrid position-error results, kilometres.

Scenario	Profile	Mean	P95	Worst	Final
Near Earth	Nominal	0.663	4.056	14.732	0.326
Near Earth	Degraded	5.408	39.255	141.305	2.974
Earth transition	Nominal	2.033	16.947	61.368	0.867
Earth transition	Degraded	16.267	167.793	593.113	5.959
Deep-space cruise	Nominal	1,103.422	2,640.408	3,952.637	1,204.435
Deep-space cruise	Degraded	168.525	412.851	749.050	235.075
Neptune capture	Nominal	0.901	3.908	14.170	0.572
Neptune capture	Degraded	6.615	36.006	129.833	4.467

Unresolved cruise anomaly. The nominal deep-space cruise full-hybrid mean error (1,103.422 km) is larger than the degraded result (168.525 km), and both are worse than their IMU-plus-pulsar counterparts. The campaign therefore exposes a scenario-specific optical-model/filter mismatch or overconfident optical weighting. Version 0.2 does not reinterpret this as successful navigation performance; it records the anomaly as a required investigation before hardware claims.

Pulsar contribution pairs full hybrid with IMU-plus-optical under identical truth and random-number order. Delta is full hybrid minus IMU-plus-optical; negative is lower with pulsar fusion. Across 160 pairs, run-mean error improved in 155, P95 in 125, and final error in 149. Worst error was unchanged in 119 pairs because a shared fault peak often dominated.

Table 4. Paired pulsar-contribution ablation. Deltas are kilometres; negative is lower with pulsar fusion.

Scenario	Profile	Delta mean	Delta P95	Delta final	Mean improved
Near Earth	Nominal	-0.683	-0.884	-0.804	20/20
Near Earth	Degraded	-4.451	-3.996	-2.991	20/20
Earth transition	Nominal	-2.376	-2.405	-2.697	20/20
Earth transition	Degraded	-23.247	-19.120	-14.856	20/20
Deep-space cruise	Nominal	-151.010	-215.112	-77.639	20/20
Deep-space cruise	Degraded	-51.146	-77.390	-71.652	20/20
Neptune capture	Nominal	-0.451	-0.648	-0.504	18/20
Neptune capture	Degraded	-2.931	-2.775	-1.562	17/20

Fault results distinguish containment from recovery. A contained fault never drove error above the defined threshold; recovered means an excursion returned below threshold for five consecutive navigation outputs; not recovered means the required sequence was absent within the bounded window. Not applicable occurs when the affected sensor is deliberately disabled by the selected suite.

Table 5. Fault outcome counts over all 640 runs.

Fault	Contained	Recovered	Not recovered	Not applicable
Camera range bias	320	0	0	320
Camera occlusion	320	0	0	320
Pulsar dropout	318	1	1	320
IMU bias excursion	623	9	8	0

6. Customer evaluation and reproducibility surface

The Customer Evaluation Console provides a local browser interface for configuring the established validation campaign without editing Rust source or a .hxns configuration file. A user can select mission windows, seed count, duration, nominal or degraded starting assumptions, processed camera/IMU/pulsar/clock uncertainties, cadence, and fault campaign. Server-side validation remains authoritative. The console invokes the existing runner in a background process and produces an HTML report, a machine-readable JSON report, run-level CSV, summary CSV, and the resolved configuration.

The console is an evaluation surface, not a second estimator. It does not modify the no_std flight core, and it does not convert marketing specifications into certified hardware performance. Its purpose is to make assumptions explicit and to let an external reviewer reproduce or challenge sensor choices, cadence, fault behaviour, and reported metrics before expensive hardware is selected.

7. Mission context

The host scenario remains a screened 2026 Earth-Jupiter-Neptune patched-conic context using the 2026-2049 DE442 pack. The retained candidate departs on 2026 day 316, reaches the Jupiter handoff on 2030 day 319, and reaches Neptune on 2046 day 068, for 19.32 years of flight. The screen reports an Earth C3 of 83.090 km^2/s^2 and a 603,186 km implied Jupiter perijove altitude, followed by an impulsive estimate for a bound 2,000 x 250,000 km-altitude Neptune orbit. These values create navigation replay windows; they are not n-body propagation, B-plane targeting, launch-site analysis, finite-burn design, or mission certification.

8. Limitations and next validation campaign

HXNS does not yet include a selected camera or pulsar instrument, detector-level processing, real sensor data, a calibrated clock model, LEON5 board drivers, reset vector, bus and interrupt handling, processor timing, radiation testing, n-body gravity-assist design, Jupiter-moon targeting, gravity harmonics, solar-radiation pressure, finite burns, or operational ground interfaces. Validation truth shares the project's mission model and ephemeris context rather than an independent orbit-determination truth source, and each bounded window cold-starts the EKF instead of carrying nineteen years of estimator state.

The next credible campaign is hardware-in-the-loop. It should replay recorded or independently generated camera and pulsar products through real interface drivers on representative LEON-class hardware; measure execution time and memory; compare HXNS with independent orbit-determination truth; investigate the cruise optical anomaly; sweep initialization, clock, bias, occultation, and catalogue errors; and configuration-control every code, sensor-model, and ephemeris version. These steps would convert transparent software evidence into an engineering validation programme.

9. Conclusion

HXNS demonstrates an embedded-oriented navigation architecture in which optical, inertial, and processed pulsar timing products are fused under explicit frame, time, cadence, uncertainty, and fault assumptions. Version 0.2 adds reproducible paired-seed evidence and a self-service evaluation surface. The results support a narrower claim than autonomous Neptune navigation: the software boundary and validation questions are concrete enough to measure, reproduce, criticize, and carry toward hardware. The retained anomaly and fault outcomes are part of that evidence, not details hidden behind the dashboard.

References

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